ABSTRACT

AIRBORNE INTERNET

Airborne Internet is an advanced communication system that provides secure and high-speed connectivity to aircraft while in flight. It uses technologies similar to the commercial internet to connect aircraft with ground stations and with other aircraft, forming a mobile and distributed network in the sky. In this system, aircraft act as network nodes that can exchange data such as weather information, flight position, and operational details, which improves flight safety and efficiency.

The system is based on standard internet protocols like TCP/IP, adapted to meet the unique requirements of the aviation environment, ensuring private, reliable, and secure communication. Airborne Internet supports various applications including in-flight Wi-Fi for passengers, real-time weather updates, flight planning, aircraft maintenance, and air-to-air data sharing for safety and security. It also plays a vital role in military operations and emergency situations by providing resilient communication when ground infrastructure is unavailable or damaged.

By reducing dependence on ground-based networks, Airborne Internet overcomes connectivity limitations over remote areas such as oceans and mountains. The technology has evolved from early aerospace communication systems, including those used by NASA, and continues to develop with the use of high-altitude platforms such as drones and balloons acting as communication relays. Overall, Airborne Internet enhances passenger experience, improves operational efficiency, increases situational awareness, and offers cost-effective communication solutions for modern aviation.

References:

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